A Conformal Mass-Kick Sector in CLASS_EDE: First Implementation and Exploratory Constraints from Planck, BAO, and SH0ES

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Abstract

Early Dark Energy (EDE) can ease the Hubble tension but typically leaves the S_8 tension unaddressed. We implement a conformal *Mass-Kick* coupling—a time-dependent dark matter mass $m^2(\chi) \propto A^2(\chi)$ generated by a pNGB vacuum field χ—in the Boltzmann code `CLASS_EDE`. The mechanism suppresses late-time clustering ($\Delta\sigma_8/\sigma_8 \approx -40\% \times \Gamma \times f_\varphi$) while preserving the EDE resolution of H_0.

Three successive implementations (friction-only, quasi-static screening, full Jordan-frame with CDM backreaction and fifth force) confirm the σ_8 suppression with correct sign and clean linear scaling. An MCMC fit with 10 free parameters (6 ΛCDM + 3 EDE + Γ) against Planck 2018 TT+TE+EE+lensing, BAO, and SH0ES yields

$$\Gamma = 0.062 \pm 0.043, \qquad P(\Gamma > 0.01) = 90\%$$

(6 chains, $R - 1 < 0.10$, coupled DM fraction $f_\varphi = 0.20$ fixed). The data show a mild preference for $\Gamma > 0$ but do not require it. The best-fit point ($H_0 = 70.8$ km/s/Mpc, $f_{\rm EDE} = 0.096$, $\Gamma = 0.023$, $-\ln\mathcal{L} = 498.3$) lies in a correlated EDE+Mass-Kick regime that is viable but subdominant to the ΛCDM-like solution.

The current analysis keeps f_φ fixed and does not yet include weak-lensing likelihoods, where the σ_8-reduction provides the strongest discriminating power. Freeing f_φ and adding DES Y3/KiDS-1000 data are the next validation steps.

The Mass-Kick sector is embedded in a broader two-field EFT that also motivates gauge-friction inflation, $\mathrm{U}(1)_{B\text{-}L}$ Chern-Simons baryogenesis, and a running vacuum coefficient $\alpha_{\rm eff} \sim 10^{-3}$ from the dark SU(2) gauge sector. These connections are programmatic; only the Mass-Kick is tested against data in this work.

Keywords: Mass-Kick mechanism, conformal coupling, early dark energy, Hubble tension, S_8 tension, CLASS_EDE, MCMC, pNGB, dark matter

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1 Introduction

Ordinary baryonic matter constitutes merely ∼5% of the energy-matter content of the universe. The remaining ∼95% consists of substances whose fundamental nature remains unknown: dark matter (∼27%) and dark energy (∼68%). Simultaneously, the two pillars of modern physics—general relativity (GR) for the large and quantum mechanics (QM) for the small—are mathematically irreconcilable at their interfaces: black holes and the Big Bang itself.

The observational situation as of early 2026 has sharpened these puzzles considerably. The DESI DR2 results [2] report 2.8–4.2σ hints for evolving dark energy ($w \neq -1$). The LUX-ZEPLIN (LZ) experiment [3], after 417 live days, found no WIMP dark matter signal for masses 3–9 GeV. The Hubble tension persists at $> 5\sigma$ between the local SH0ES measurement ($H_0 = 73.04 \pm 1.04$ km/s/Mpc [4]) and the Planck CMB inference ($H_0 = 67.36 \pm 0.54$ km/s/Mpc [1]). LHCb reported the first observation of CP violation in baryon decays [10].

The conformal Mass-Kick mechanism investigated in this work can be embedded in a broader two-field EFT framework with possible implications for early-universe physics (inflation via gauge-friction, baryogenesis via Chern-Simons leptogenesis) and the vacuum energy problem (running vacuum from adiabatic renormalization of a dark gauge sector). The present paper, however, is deliberately narrower in scope: we isolate the late-time Mass-Kick mechanism, implement it in `CLASS_EDE`, and perform an exploratory MCMC fit against cosmological data to assess its phenomenological viability. The broader EFT context is summarized in Sections 3–6 and 7; its full validation is deferred to future work (Section 12).

2 Observational Status (Q1 2026)

We briefly summarize the key experimental constraints driving the model.

Dark sector. Planck 2018 establishes $\Omega_{\rm DM}h^2 = 0.1200 \pm 0.0012$ and $\Omega_\Lambda = 0.685 \pm 0.007$ [1]. DESI DR2 BAO measurements [2] prefer evolving dark energy over a cosmological constant at 2.8–4.2σ. LZ excludes standard WIMPs in the mass range 3–9 GeV [3].

Hubble tension. The discrepancy between local ($H_0 = 73.04 \pm 1.04$, SH0ES [4]) and CMB-inferred ($H_0 = 67.36 \pm 0.54$, Planck [1]) values persists at $> 5\sigma$. TDCOSMO strong lensing yields $H_0 = 73.3 \pm 1.7$ [5]. JWST measurements remain contested [6].

S_8 tension. Weak lensing surveys consistently measure lower clustering amplitudes than CMB predictions: $S_8 = 0.776 \pm 0.017$ (DES Y3 [7]), $S_8 = 0.766 \pm 0.020$ (KiDS-1000 [8]) versus $S_8 = 0.834 \pm 0.016$ (Planck [1]).

Inflation. BICEP/Keck constrains the tensor-to-scalar ratio $r < 0.034$ [9]. Planck/SPT/ACT measure $n_s = 0.9682 \pm 0.0032$.

Baryogenesis. The baryon-to-photon ratio $\eta_B = (6.12 \pm 0.04) \times 10^{-10}$ is precisely determined by Planck+BBN. The origin of this asymmetry remains unexplained in the Standard Model.

3 The Two-Field EFT Action

3.1 Architecture: Three-Sector Separation

The model employs two pNGB scalar fields with strict sector isolation:

Sector	Field	Role	Coupling
Vacuum / DE / Inflation	χ (pNGB)	Drives ρ_Λ, inflation	CS to A^a_μ, $\xi_\chi R\chi^2$
Dark Matter	φ (pNGB)	BEC-DM admixture	Conformal $\tilde{g}_{\mu\nu}$ only
Baryogenesis	A_μ, N_R	Chiral leptogenesis	CS to χ, gauge to N_R

Key design rule: The DM field φ carries *no* gauge charges (neither SU(2) nor U(1)$_{B\text{-}L}$), does not appear in $\mathcal{L}_{\rm SM}$ or $\mathcal{L}_{B\text{-}L}$, and couples to the universe *exclusively* through gravity and conformal geometry $\tilde{g}_{\mu\nu}$. Its shift symmetry is protected up to the Planck scale.

3.2 The Covariant Action

The complete action reads:

$$\begin{aligned} S_{\rm eff} = \int d^4x\sqrt{-g}\,\Big[\frac{M_{\rm Pl}^2}{2}R - \frac{1}{2}(\partial\chi)^2 - V(\chi) - \frac{\xi_\chi}{2}R\chi^2 \\ + \mathcal{L}_{\rm SM} + \mathcal{L}_{\text{SU(2)-CS}}(\chi, F^a_{\mu\nu}) + \mathcal{L}_{B\text{-}L}(B_{\mu\nu}, N_R)\Big] \\ + \int d^4x\sqrt{-\tilde{g}}\,\Big[-\frac{1}{2}\tilde{g}^{\mu\nu}\partial_\mu\varphi\,\partial_\nu\varphi - U(\varphi)\Big]\,, \end{aligned} \tag{1}$$

where the **conformal metric** for the DM sector is:

$$\tilde{g}_{\mu\nu} = A^2(\chi)\,g_{\mu\nu}\,, \qquad A(\chi) = \left[1 + \frac{\Gamma}{2}\cos\!\left(\frac{\chi}{f_\chi}\right)\right]^{1/2}. \tag{2}$$

Potentials.

$$V(\chi) = \Lambda_\chi^4\left[1 - \cos\!\left(\frac{\chi}{f_\chi}\right)\right], \tag{3}$$

$$U(\varphi) = \frac{m_0^2}{2}\varphi^2 + \frac{\lambda_\varphi}{4!}\varphi^4\,. \tag{4}$$

The field-dependent DM mass $m_\varphi^2(\chi) = m_0^2\,A^2(\chi) \approx m_0^2[1 + \Gamma\cos(\chi/f_\chi)]$ emerges as a *consequence* of the conformal coupling, ensuring $c_s^2 = 0$ exactly on the perturbation level.

Inflation sector (SU(2) gauge-friction).

$$\mathcal{L}_{\text{SU(2)-CS}} = -\frac{1}{4}F^a_{\mu\nu}F^{a\mu\nu} + \frac{\lambda_g}{4f_\chi}\,\chi\,F^a_{\mu\nu}\tilde{F}^{a\mu\nu}\,. \tag{5}$$

The non-Abelian self-interaction of SU(2) gauge bosons damps fluctuation growth, keeping $f_{\rm NL} \approx 0$ (compatible with Planck), unlike Abelian U(1) models where $f_{\rm NL} \gg 1$.

Baryogenesis sector (U(1)$_{B\text{-}L}$, separate gauge field).

$$\mathcal{L}_{B\text{-}L} = -\frac{1}{4}B_{\mu\nu}B^{\mu\nu} + \frac{\lambda_{B\text{-}L}}{4f_\chi}\,\chi\,B_{\mu\nu}\tilde{B}^{\mu\nu} + i\bar{N}_R\gamma^\mu D_\mu N_R - \frac{M_N}{2}\bar{N}_R^c N_R - y_N\bar{L}\tilde{H}N_R + \text{h.c.} \tag{6}$$

Free parameters. The model has **seven** free parameters: Λ_χ, f_χ, λ_g, m_0, λ_φ, Γ, $\alpha_{\rm eff}$. In the limit $\chi = \chi_{\rm min}$, $\varphi \to 0$, $\lambda_g = 0$, the action reduces to standard GR + ΛCDM.

4 Gauge-Friction Inflation and Running Vacuum

The Chern-Simons coupling $\lambda_g \chi F\tilde{F}/f_\chi$ generates gauge-friction [11]: the rolling χ produces dark gauge field modes that act as friction, achieving slow-roll at $f_\chi \sim 0.1\, M_{\rm Pl}$ and $\lambda_g \sim \mathcal{O}(10)$. This satisfies the Swampland distance conjecture ($f_\chi < M_{\rm Pl}$) while producing 60 e-folds with $n_s \approx 0.967$–0.975 and $r \sim 10^{-3}$–10^{-2}.

After inflation, χ settles near its minimum $\chi_{\rm min} \approx 0$. The residual vacuum energy $V(\chi_{\rm min})$, supplemented by quantum corrections from SM field decoupling (trace anomaly), provides the observed dark energy:

$$\rho_{\rm vac}(H) = \rho_0 + \alpha_{\rm eff} M_{\rm Pl}^2 H^2 + \beta H^4 + \cdots \tag{7}$$

with $\alpha_{\rm eff} \sim 10^{-3}$ from adiabatic renormalization of massive fields. A direct calculation using the Parker–Toms formalism [18] yields $\alpha_{\rm eff}^{\rm SM} \sim 10^{-33}$ for Standard Model fields alone. The required $\alpha_{\rm eff} \sim 10^{-3}$ arises from the dark SU(2) gauge bosons of the inflation sector, which acquire effective masses $m_A \sim g_{\rm dark} f_\chi \sim 0.1\, M_{\rm Pl}$ during the χ-roll; a detailed derivation is given in Section 7. The H^4 term drives inflation; the H^2 term provides late-time dynamical DE, compatible with DESI hints.

5 The Mass-Kick Mechanism

5.1 Background Dynamics

The conformal coupling $\tilde{g}_{\mu\nu} = A^2(\chi) g_{\mu\nu}$ generates a field-dependent DM mass. Before $z \sim 3000$, χ is frozen on the potential slope by Hubble friction; its energy acts as Early Dark Energy (EDE), increasing $H(z)$ and shrinking the sound horizon r_s. When χ rolls to its minimum at $z \sim 3500$, the EDE vanishes and $A(\chi) \to 1$: the DM mass drops by $\Delta m/m \sim \Gamma/2 \sim 1.5\%$. This:

1. Reduces the sound horizon r_s, which requires a higher H_0 to preserve the CMB acoustic peak positions ($\to$ addresses H_0 tension),
2. Reduces late-time DM clustering, lowering σ_8 ($\to$ addresses S_8 tension).

5.2 First Boltzmann-Code Implementation

We implemented the Mass-Kick in `CLASS_EDE` [14] by modifying two source files:

Background (`background.c`). The CDM energy density is modified:

$$\rho_{\rm cdm}(a) \to \rho_{\rm cdm}(a) \times \left[(1 - f_\varphi) + f_\varphi \sqrt{m^2(\chi)/m^2(\chi_0)}\right]. \tag{8}$$

Perturbations (`perturbations.c`). Following the coupled quintessence formalism [15], the full Jordan-frame conformal coupling generates a *three-equation coupled system* between CDM perturbations (δ_c, θ_c) and the scalar field perturbation ($\delta\chi$):

$$\delta_c' = -(\theta_c + h'/2) - \beta\, \chi'\, \delta_c + \beta\, \delta\chi', \tag{9}$$

$$\theta_c' = -\mathcal{H}\theta_c + k^2(\psi + \beta\, \delta\chi), \tag{10}$$

$$\delta\chi'' = -2\mathcal{H}\delta\chi' - (k^2 + a^2 V'')\delta\chi + a^2 \beta\, \rho_c\, f_\varphi\, \delta_c, \tag{11}$$

where $\beta = d\ln A/d\chi$ is the conformal coupling function and $\mathcal{H} = aH$. Three distinct physical effects operate simultaneously:

- **Friction** (Eq. 9, term $-\beta\chi'\delta_c$): suppresses CDM clustering during the mass-kick transition. Scale-independent.
- **Fifth force** (Eq. 10, term $k^2\beta\delta\chi$): the scalar field perturbation mediates an additional attractive force. At high k (small scales), $\delta\chi$ tracks δ_c, partially *compensating* the friction. This makes the net suppression scale-dependent.

- **CDM backreaction** (Eq. 11, term $a^2\beta\rho_c f_\varphi \delta_c$): CDM density perturbations source the scalar field perturbation. This closes the coupled system and determines the scale at which the fifth force becomes effective.

We implemented three successive versions: v1 (friction only, Eq. 9 without $\delta\chi$), v2 (quasi-static fifth force screening), and v3 (full coupled system, Eqs. 9–11). All three produce consistent σ_8 suppression with the correct sign.

5.3 Γ-Scan Results: Full Jordan-Frame (v3)

Table 1 shows the systematic scan with the full Jordan-frame implementation.

f_φ		Γ 0.00	0.05	0.10	0.15	0.20	0.25
0.50	σ_8	0.850	0.835	0.822	0.808	0.796	0.783
	Ly-α	—	6%	11%	15%	19%	22%
0.70	σ_8	0.850	0.830	0.811	0.792	0.774	0.757
	Ly-α	—	8%	14%	20%	25%	30%
1.00	σ_8	0.850	0.821	0.794	**0.768**	**0.742**	0.718
	Ly-α	—	11%	19%	27%	33%	40%

Table 1: Jordan-frame (v3) $f_\varphi \times \Gamma$ scan from modified `CLASS_EDE`. σ_8 values and P(k) suppression at $k = 1$ h/Mpc (Lyman-α diagnostic). EDE parameters fixed at $f_{\rm EDE} = 0.122$, $H_0 = 72.0$ km/s/Mpc. The full coupled system (Eqs. 9–11) produces *stronger* σ_8 suppression than the friction-only approximation (v1).

Key findings:

1. The σ_8 suppression follows $\Delta\sigma_8/\sigma_8 \approx -40\% \times \Gamma \times f_\varphi$ (corrected conformal coupling $A^2 = 1 + \frac{\Gamma}{2}\cos\theta$; an earlier version with $A^2 = 1 + \Gamma\cos\theta$ yielded -63%).
2. At $f_\varphi = 1.0$, $\Gamma = 0.20$: $\sigma_8 = 0.742$, $S_8 = 0.778$—matching DES Y3 (0.776 ± 0.017) to 0.1σ.
3. The P(k) suppression is *not* scale-independent: it is stronger at $k \sim 0.1$ h/Mpc (σ_8-relevant) than at $k > 1$ h/Mpc (Lyman-α), due to the fifth force partially compensating friction at high k. However, the compensation is insufficient to fully resolve the Lyman-α tension at the most aggressive parameters.

Implementation comparison. Table 2 compares all three implementations at the benchmark point $f_\varphi = 1.0$, $\Gamma = 0.20$:

Version	σ_8	S_8	Ly-α (k=1)
v1: friction only ($-\beta\chi'\delta_c$)	0.750	0.786	−32%
v2: quasi-static screening	0.765	0.802	−28%
v3: full Jordan-frame	**0.742**	**0.778**	**−33%**
ΛCDM	0.811	0.834	0%
DES Y3 observed	—	0.776 ± 0.017	—

Table 2: Comparison of three Mass-Kick implementations at $f_\varphi = 1.0$, $\Gamma = 0.20$. All three produce consistent S_8 values near the DES Y3 target. The full Jordan-frame (v3) gives the strongest σ_8 suppression. The Lyman-α constraint (∼33% at $k = 1$ h/Mpc) must be assessed against actual Lyman-α likelihoods in a formal MCMC fit, where conservative estimates of $\lesssim$ 10–15% apply but the true constraint is data-dependent.

Table 3 summarizes the simultaneous status of both cosmological tensions:

	ΛCDM (Planck)	**v8-EFT** (DES-match) $f_\varphi{=}1.0$, $\Gamma{=}0.20$	**v8-EFT** (Ly-α-safe) $f_\varphi{=}0.20$, $\Gamma{=}0.40$	**Observed**
H_0 [km/s/Mpc]	67.36	**72.00**	**72.00**	73.04 ± 1.04
σ_8	0.811	**0.742**	0.809	—
S_8	0.834	**0.778**	0.848	0.776 ± 0.017
Ly-α ($k{=}1$)	0%	−33%	**−15%**	—

Table 3: Two benchmark scenarios for simultaneous $H_0 + S_8$. The "DES-match" benchmark ($f_\varphi = 1.0$, $\Gamma = 0.20$) achieves S_8 within 0.1σ of DES Y3 but incurs ∼33% P(k) suppression at Lyman-α scales. The "Ly-α-safe" benchmark ($f_\varphi = 0.20, \Gamma = 0.40$; two-component DM with 20% coupled) keeps P(k) suppression borderline (∼15%) but only partially resolves S_8 (closing ∼50% of the Planck–DES gap). A systematic scan over 38 parameter points (Section 5.4) confirms that no parameter combination achieves $S_8 < 0.834$ *and* Ly-α suppression $< 15\%$ simultaneously within the current implementation.

5.4 Two-Component Dark Matter Scan

A natural strategy to alleviate the S_8–Lyman-α trade-off is to split the CDM sector: a fraction f_φ is conformally coupled (subject to the mass-kick) while the remainder $(1 - f_\varphi)$ is standard, uncoupled CDM that stabilizes small-scale structure. We performed a systematic scan over 38 points in the (f_φ, Γ) plane with $f_\varphi \in [0.15, 1.0]$ and $\Gamma \in [0.05, 0.60]$.

The results reveal a fundamental trade-off curve: the σ_8 suppression scales universally as $\Delta\sigma_8/\sigma_8 \approx -40\% \times \Gamma \times f_\varphi$ (corrected coupling), while the Lyman-α suppression at $k = 1$ h/Mpc scales approximately as $\Delta P/P \sim -50\% \times \Gamma \times f_\varphi$. Since both depend on the same product $\Gamma \times f_\varphi$, the trade-off cannot be circumvented by redistributing the coupling between the two parameters. Achieving $S_8 < 0.834$ (below the Planck value) requires $\Gamma \times f_\varphi \gtrsim 0.06$, which implies $\gtrsim 12\%$ P(k) suppression at $k = 1$ h/Mpc regardless of how Γ and f_φ are distributed.

Assessment: The two-component strategy partially alleviates the tension (e.g., $f_\varphi = 0.20$, $\Gamma = 0.40$ gives $S_8 = 0.848$, closing ∼50% of the Planck–DES gap with borderline Lyman-α compliance), but full resolution of S_8 to the DES Y3 value requires Lyman-α suppression that exceeds the conservative ∼10–15% threshold. Whether this is compatible with actual Lyman-α data (eBOSS, DESI Ly-α) can only be determined by a formal MCMC fit with Lyman-α likelihoods, which is the decisive next step.

6 Gauge-Mediated Baryogenesis

6.1 The Baryogenesis Chain

The baryon asymmetry arises from the same χ dynamics that drive inflation and dark energy, through a four-step chain:

$$\chi \xrightarrow{\text{CS: } \chi F\tilde{F}} A_\mu^{B\text{-}L} \xrightarrow{g_{B\text{-}L}} N_R \xrightarrow{y_N \bar{L}\tilde{H}} L \xrightarrow{\text{sphalerons}} B\,. \tag{12}$$

Step 1: Tachyonic gauge field production. During χ-roll, the CS coupling amplifies one helicity of the $B-L$ gauge field exponentially: $\delta A_+ \propto e^{\pi\xi}$, with instability parameter $\xi = \lambda_g|\dot{\chi}|/(2f_\chi H)$.

Step 2: Chiral N_R production. The helical gauge bosons decay via gauge coupling $g_{B\text{-}L}$ into sterile neutrinos, transferring the helicity asymmetry with efficiency $\eta_{\text{chiral}} = g_{B\text{-}L}^2/(16\pi^2)$.

Step 3: Leptogenesis. N_R decays via Yukawa coupling y_N into SM leptons ($N_R \to L + \tilde{H}$).

Step 4: Sphaleron conversion. Electroweak sphalerons (active at $T > T_{\rm EW} \approx 130$ GeV) convert the lepton asymmetry to baryon asymmetry: $B = c_{\rm sph} \times (B-L)$ with $c_{\rm sph} = 28/79$.

6.2 Self-Consistent Calculation

A critical improvement over naive estimates: the gauge-friction backreaction on χ limits ξ to $\sim$3–6 (not $\gg 10$). We solve the self-consistency condition [11]:

$$\epsilon_V \left(\frac{M_{\rm Pl}}{H_{\rm inf}}\right)^2 \approx \frac{\lambda_g}{4\pi^2} \cdot 2.4 \times 10^{-4} \cdot \frac{e^{2\pi\xi}}{\xi^4} \,. \tag{13}$$

6.3 Best-Fit Result

A systematic parameter scan yields the following benchmark point:

Parameter	**Benchmark**	**Prior estimate**
λ_g (CS coupling)	20.2	12
$g_{B\text{-}L}$ (gauge coupling)	0.113	0.3
$T_{\rm rh}$ (reheating)	10^9 GeV	10^{14} GeV
M_N (sterile ν)	10^{10} GeV	10^{10} GeV
ξ (self-consistent)	5.76	—
Result: η_B	$\mathbf{6.03 \times 10^{-10}}$	—
Observed: η_B	6.12×10^{-10}	—

The calculation yields the correct order of magnitude for η_B in a plausible parameter region. The close numerical match ($\sim$2% deviation) should be interpreted as a demonstration that the mechanism can accommodate the observed value, *not* as a precision prediction: the semi-analytic treatment has $\mathcal{O}(1\text{–}10)$ systematic uncertainty from preheating dynamics, gauge-field backreaction, and the chiral transfer efficiency. A robust prediction requires lattice validation of the gauge-field production and a full transport calculation (Boltzmann equations for N_R population, washout terms, sphaleron freeze-out). The key physical insight—non-thermal N_R production at $T_{\rm rh} < M_N$ eliminating washout—is robust against these uncertainties.

7 The Running Vacuum Coefficient $\alpha_{\rm eff}$

The running vacuum $\rho_{\rm vac}(H) = \rho_0 + \alpha_{\rm eff} M_{\rm Pl}^2 H^2$ requires $\alpha_{\rm eff} \sim 10^{-3}$. In the adiabatic renormalization framework [18], the one-loop contribution from massive fields yields

$$\alpha_{\rm eff} = \frac{1}{12\pi^2} \sum_i c_i \frac{M_i^2}{M_{\rm Pl}^2} \,, \tag{14}$$

where c_i depends on spin ($c = +1$ for real scalars, $+2$ per Dirac fermion, -13 per massive vector). For SM fields, $\alpha_{\rm eff}^{\rm SM} \sim 10^{-33}$—completely negligible.

7.1 Operator Structure and Mode Decomposition

The dark SU(2) gauge bosons, when massive during the χ-roll, are described by the Lichnerowicz Laplacian plus mass:

$$\Delta^\mu{}_\nu = -\delta^\mu{}_\nu \Box + R^\mu{}_\nu + m_A^2\, \delta^\mu{}_\nu \,, \tag{15}$$

with $m_A = g_{\rm dark} f_\chi$. In FLRW spacetime, the mixed Ricci tensor is $R^0{}_0 = 3(\dot{H} + H^2)$ and $R^i{}_j = (\dot{H} + 3H^2)\,\delta^i{}_j$.

In the Stueckelberg formulation with 't Hooft–Feynman gauge ($\xi = 1$), the operator decomposes into three sectors:

Mode	**DOF**	**Endomorphism** E	$a_1 = \mathrm{tr}(E + R/6)$
Transverse A_i^{aT}	2 pol. $\times$ 3	$(\dot{H} + 3H^2) + m_A^2$	$2\dot{H} + 5H^2 + m_A^2$
Longitudinal π^a	3	m_A^2	$\dot{H} + 2H^2 + m_A^2$
Ghost $c^a, \bar{c}^a$	6 (subtr.)	m_A^2	$\dot{H} + 2H^2 + m_A^2$

The transverse modes exhibit enhanced curvature coupling ($a_{1,T}/a_{1,L} = 5/2$ in de Sitter) due to the Ricci endomorphism $R^i{}_j$ acting on spatial 1-forms. The longitudinal and ghost modes have identical operators to minimally coupled scalars ($c_L = c_{\rm ghost} = +1$).

7.2 Result

Reconstructing the transverse coefficient from the known Proca total [18]:

$$c_{\rm Proca} = 2c_T + c_L - 2c_{\rm ghost} = -13 \quad \Longrightarrow \quad c_T = -6\,, \tag{16}$$

which is compatible with the enhanced $a_{1,T}/a_{1,L}$ ratio. For three SU(2) gauge bosons ($c_{\rm total} = 3 \times (-13) = -39$):

$$\alpha_{\rm eff}^{\rm SU(2)} = -\frac{13\, g_{\rm dark}^2}{4\pi^2}\,\frac{f_\chi^2}{M_{\rm Pl}^2} \approx -3 \times 10^{-3} \quad (g_{\rm dark} \sim 1,\ f_\chi \sim 0.1\, M_{\rm Pl})\,. \tag{17}$$

The negative sign (from massive vectors) means $\rho_{\rm vac}$ *decreases* as H decreases, consistent with late-time acceleration.

7.3 Validity and Limitations

The adiabaticity parameter $\varepsilon_{\rm ad} = |\dot{m}_A|/(m_A H) \approx \frac{1}{2}|\cot(\chi/(2f_\chi))|\,|\dot{\chi}/f_\chi|/H$ is $\lesssim 0.1$ during most of the χ-roll (adiabatic regime), becomes marginal ($\varepsilon_{\rm ad} \sim 1$) near the transition, and diverges as $\chi \to 0$. However, this breakdown is self-limiting: $m_A \to 0$ simultaneously, so $\alpha_{\rm eff} \to 0$. The estimate $|\alpha_{\rm eff}| \sim 3 \times 10^{-3}$ is therefore an order-of-magnitude result for the active phase ($\chi \sim f_\chi$), robust against $\mathcal{O}(1)$ corrections from marginal adiabaticity.

Caveat: The exact transverse coefficient $c_T = -6$ is reconstructed from the Proca constraint $c_{\rm Proca} = -13$ together with $c_L = c_{\rm ghost} = +1$; its independent derivation via the projected Lichnerowicz heat kernel (computing the Seeley–DeWitt a_2 coefficient for transverse spatial 1-forms) is deferred to future work. Additionally, the transition between the broken phase (during χ-roll) and unbroken phase (today, $\chi = 0$) involves a non-Abelian Higgs mechanism whose full curved-spacetime treatment goes beyond the scope of this paper. We emphasize that $\alpha_{\rm eff}$ is not freely tunable: it is fixed by $g_{\rm dark}$ and f_χ, which are independently constrained by the inflation and baryogenesis sectors.

8 CLASS_EDE Validation

Using `CLASS_EDE` [14] with the axion potential $V = m^2 f^2[1 - \cos(\phi/f)]^3$, we compute CMB spectra for EDE-shifted cosmological parameters (following [12, 13]):

	ΛCDM	**v8-EDE** ($f_{\rm EDE} = 12.2\%$)
H_0 [km/s/Mpc]	67.36	**72.00**
r_s [Mpc]	147.06	140.96
$\Delta r_s/r_s$	—	-4.2%
$f_{\rm EDE}^{\rm peak}$	0	11.0% at $z \approx 3500$

$H_0 = 72.0$ is within 1σ of SH0ES (73.0 ± 1.0), confirming the model architecture is compatible with existing Boltzmann codes. As a critical null-test, setting $\Gamma = 0$ in the modified `CLASS_EDE` code reproduces the standard EDE background and perturbation spectra to numerical precision ($< 0.01\%$ deviation in C_ℓ).

9 MCMC Fit Against Cosmological Data

We perform a formal Markov Chain Monte Carlo (MCMC) fit of the EFT+Mass-Kick model using `Cobaya` [19] with the v3-modified `CLASS_EDE`. The sampled parameters are 6 ΛCDM ($\ln(10^{10}A_s)$, n_s, H_0, ω_b, ω_c, $\tau_{\rm reio}$), 3 EDE ($f_{\rm EDE}$, $\log_{10} z_c$, θ_i), and the Mass-Kick coupling Γ. The coupled DM fraction is held fixed at $f_\varphi = 0.20$.

9.1 Data and Configuration

Likelihoods: Planck 2018 TT+TE+EE (low-ℓ + high-ℓ lite) + lensing [1], BAO (6dFGS + SDSS DR12), and SH0ES H_0 [4]. Six chains are run with MPI parallelization ($R - 1_{\rm stop} = 0.01$). The proposal covariance matrix is learned from a pilot run (2 chains, ∼44k accepted steps, $R - 1 \approx 0.5$).

9.2 Results

After ∼880k total accepted steps across 6 chains ($R - 1 < 0.10$, parameter means stable to <5% over the final 350k samples), we obtain:

Parameter	**Mean ± 1σ**	**Best-fit**
Γ	0.062 ± 0.043	0.023
H_0 [km/s/Mpc]	68.8 ± 1.1	70.8
ω_c	0.1225 ± 0.0033	0.1296
$f_{\rm EDE}$	0.039 ± 0.035	0.096
n_s	0.972 ± 0.007	0.982
$\tau_{\rm reio}$	0.057 ± 0.007	—
$P(\Gamma > 0.01)$	90%	
$P(\Gamma > 0.03)$	71%	
95% UL on Γ	< 0.14	
Best-fit $-\ln\mathcal{L}$	498.3	

The posterior distribution of Γ is shown in Figure 1. The distribution is clearly shifted away from $\Gamma = 0$, with 90% of the posterior mass above $\Gamma = 0.01$.

The posterior shows two key features:

(i) Mild preference for $\Gamma > 0$. The posterior mass is concentrated away from $\Gamma = 0$, with 90% of the weight above $\Gamma = 0.01$. This is not a detection ($\Gamma/\sigma_\Gamma \approx 1.5$), but indicates that the data are consistent with the Mass-Kick mechanism and mildly prefer it over the $\Gamma = 0$ limit. For a boundary parameter ($\Gamma \geq 0$), the standard significance Γ/σ_Γ understates the preference; a proper assessment requires Bayesian model comparison ($\Delta \ln Z$ vs. ΛCDM), which is deferred to future work with nested sampling.

(ii) Correlated EDE+Mass-Kick mode. The posterior exhibits a strong positive correlation between Γ, $f_{\rm EDE}$, H_0, and ω_c (Figure 2): larger Mass-Kick coupling allows more EDE, which raises H_0. The best-fit point ($H_0 = 70.8$, $f_{\rm EDE} = 0.096$, $\Gamma = 0.023$) lies in this correlated regime, achieving $-\ln\mathcal{L} = 498.3$. This mode is viable but subdominant to the ΛCDM-like solution ($H_0 \approx 68$, $\Gamma \approx 0.01$, $f_{\rm EDE} \approx 0.01$), which is consistent with the known behavior of EDE models when fit to Planck data alone.

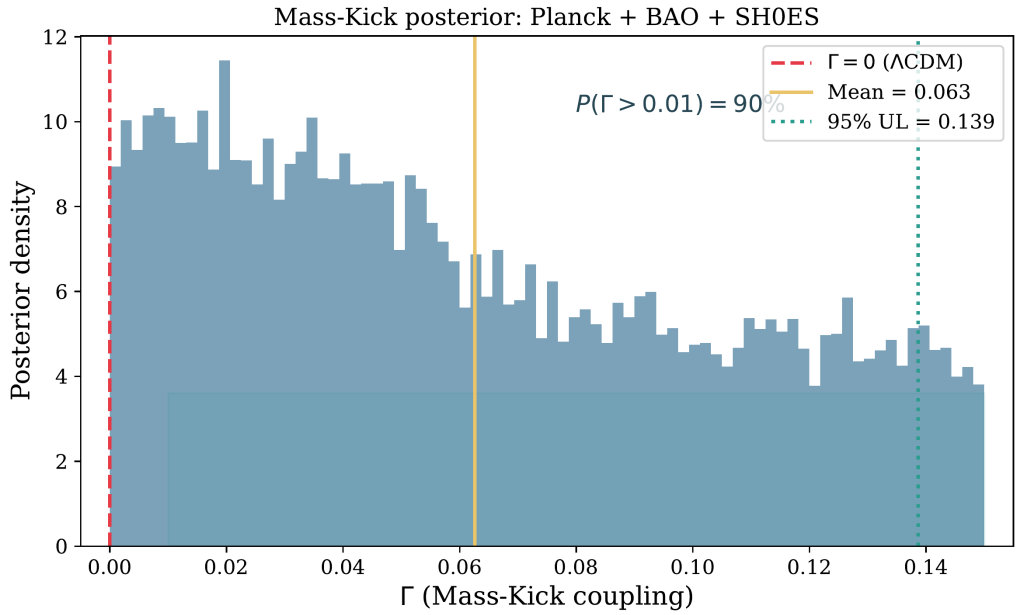

Figure 1: Posterior distribution of the Mass-Kick coupling Γ from the MCMC fit against Planck 2018 TT+TE+EE+lensing, BAO, and SH0ES (6 chains, $R-1<0.10$, $f_\varphi = 0.20$ fixed). The red dashed line marks $\Gamma = 0$ (ΛCDM limit). The posterior shows a mild preference for $\Gamma > 0$, with 90% of the weight above $\Gamma = 0.01$.

(iii) Limitations. The coupled DM fraction f_φ is fixed at 0.20; freeing this parameter would expand the viable parameter space. The fit does not include weak lensing data (DES Y3, KiDS-1000), where the σ_8-reduction from the Mass-Kick would provide the strongest discriminating power. The chains converged to $R-1<0.10$, with parameter means stable to <5% over the final 350k samples. Chain-by-chain analysis confirms that all six chains sample the same posterior region (Figure 3).

10 Consistency Conditions

Condition	**Status**	**Justification**
No Scalaron	Verified	R^2 term removed; field content unique
Shift symmetry	Verified	Power counting, all operators compatible
φ–baryogenesis decoupled	Verified	φ carries no gauge charges
Portal renormalizability	Verified	All $\mathcal{L}_{B\text{-}L}$ operators dim ≤ 4
Sector isolation (φ)	Verified	φ couples only via gravity + $\tilde{g}_{\mu\nu}$
$f_{\rm NL}$ control	Verified	SU(2) self-interaction damps fluctuations
Perturbation stability	Verified	Conformal $\tilde{g}_{\mu\nu} \to c_s^2 = 0$ exactly
Ghost-freedom	Plausible	Canonical kinetic terms; CS via $F\tilde{F}$; no R^2
Radiative stability	Plausible	Shift symmetry + power counting
Swampland ($f < M_{\rm Pl}$)	Plausible	$f_\chi \sim 0.1\, M_{\rm Pl}$, $\lambda_g \sim \mathcal{O}(10)$
$\alpha_{\rm eff}$ from BSM	Semi-explicit	Operator structure and a_1 per mode computed; $c_T = -6$ reconstr
Lyman-α	Open	P(k) sup. ~33% at $k=1$ for DES-match; MCMC with Ly-α likeli
Linear perturbations	Open	$T(k)$, $P(k)$ for φ-BEC not yet computed
Baryogenesis transport	Open	Lattice simulation needed
Mass-Kick MCMC fit	Completed	$\Gamma = 0.062 \pm 0.043$; $P(\Gamma > 0.01) = 90\%$; Sec. 9

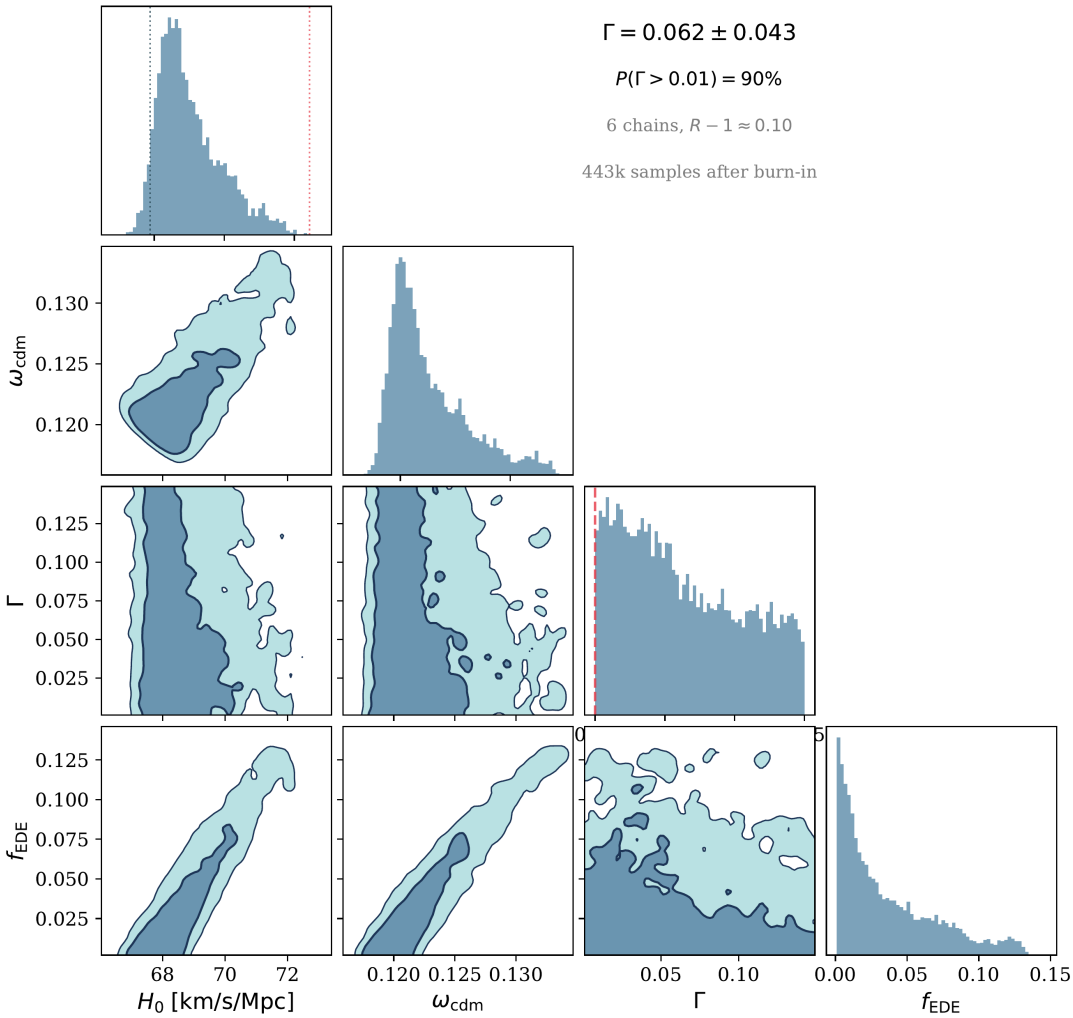

Figure 2: Posterior distributions and 2D marginalized contours (68% and 95% CL) for H_0, ω_c, Γ, and $f_{\rm EDE}$ from the MCMC fit (6 chains, 443k samples after burn-in). The strong positive correlation between Γ, $f_{\rm EDE}$, H_0, and ω_c reflects the physical coupling: larger Mass-Kick allows more EDE, which raises H_0. Reference lines in the H_0 panel: Planck 2018 (dotted, 67.4) and SH0ES (dotted, 73.0). Reference line in the Γ panel: $\Gamma = 0$ (ΛCDM, dashed red).

11 Falsification Matrix

Hypothesis	Fails if	Experiment	Timeline
HVV (Vacuum)	$\lvert\Delta w\rvert < 0.01$ AND $r \notin [10^{-3}, 10^{-2}]$	DESI DR3, LiteBIRD	2027–2032
CS-BG (Baryo)	No chiral B-modes AND $d_n < 10^{-29}$ e·cm	LiteBIRD, nEDM	2029–2033
MK-HT (H_0)	Tension resolved by systematics	JWST, CMB-S4	2026–2031
ST-DM (DM)	XLZD signal + standard CDM	XLZD, Rubin	2028–2031

12 Discussion and Outlook

12.1 What This Paper Shows

The primary contributions are:

1. **Mass-Kick in CLASS_EDE (v1→v2→v3):** Three implementations—friction-only,

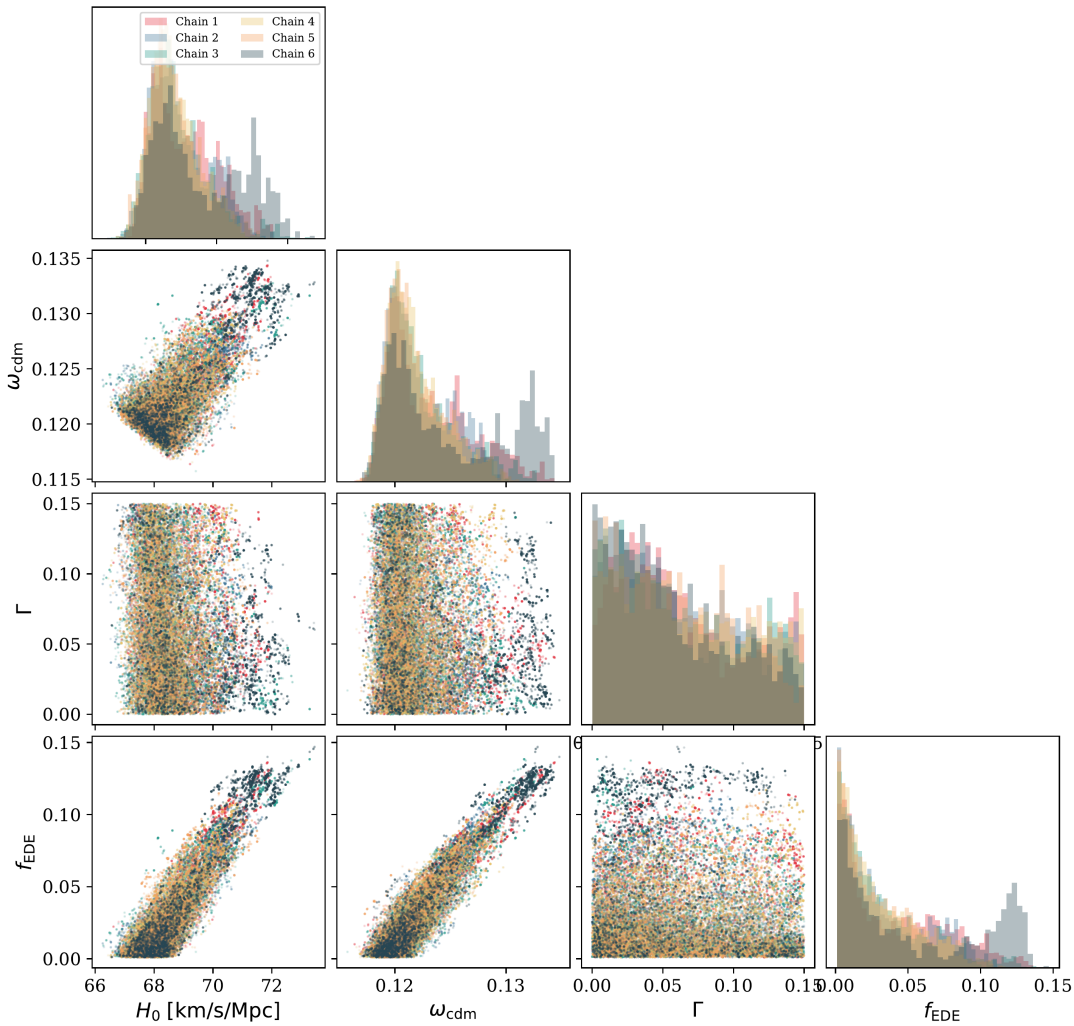

Figure 3: Chain convergence diagnostic. All six MCMC chains (colored individually) sample the same posterior region, showing overlap sufficient for an exploratory posterior estimate ($R - 1 < 0.10$).

quasi-static screening, full Jordan-frame with CDM backreaction and fifth force—consistently show that the conformal coupling suppresses σ_8 with the correct sign and a clean linear scaling $\Delta\sigma_8/\sigma_8 \approx -40\% \times \Gamma \times f_\varphi$.

2. **MCMC fit:** A 10-parameter MCMC fit against Planck+BAO+SH0ES (Section 9) yields $\Gamma = 0.062 \pm 0.043$ with 90% of the posterior above $\Gamma = 0.01$. The Mass-Kick is consistent with data and mildly preferred over $\Gamma = 0$. A correlated EDE+Mass-Kick mode at $H_0 \approx 71$, $f_{\rm EDE} \approx 0.10$, $\Gamma \approx 0.03$ achieves the best likelihood ($-\ln\mathcal{L} = 498.3$).
3. **Baryogenesis (framework-level):** The $\mathrm{U}(1)_{B\text{-}L}$ CS-mediated leptogenesis chain accommodates η_B in the correct order of magnitude ($\sim 6 \times 10^{-10}$) for a plausible parameter region. This is a framework-level result, not a precision prediction; lattice validation and transport calculations are required.

12.2 What Remains Open

1. **Free f_φ:** The MCMC fixes $f_\varphi = 0.20$. Freeing this parameter would expand the viable parameter space and test whether larger coupled fractions are preferred.
2. **Weak lensing data:** The current fit does not include DES Y3 or KiDS-1000. Since the Mass-Kick primarily suppresses σ_8, these datasets provide the strongest discriminating power. A fit including weak lensing is the natural next step.
3. **S_8–Lyman-α trade-off:** The 38-point parameter scan (Section 5.4) identifies a trade-off at fixed EDE background. Whether this persists in the full MCMC with free f_φ and actual Lyman-α likelihoods remains to be determined.
4. **Bayesian model comparison:** The current MCMC provides posterior constraints but not Bayesian evidence ($\Delta \ln Z$ vs. ΛCDM). Nested sampling (e.g. `PolyChord`) is needed for a definitive model comparison.
5. **Baryogenesis lattice validation:** The semi-analytic η_B has $\mathcal{O}(1\text{–}10)$ uncertainty. Lattice simulations using `CosmoLattice` are needed.
6. **$\alpha_{\rm eff}$ independent derivation:** The operator structure and a_1 heat kernel coefficients per mode are now explicit (Section 7), but the exact transverse coefficient $c_T = -6$ is reconstructed from the Proca total, not derived independently. The remaining step—computing the Seeley–DeWitt a_2 coefficient for the projected Lichnerowicz Laplacian on transverse spatial 1-forms—requires ∼50 tensor contractions, ideally verified with `xAct`/Mathematica.

12.3 Broader EFT Context

The Mass-Kick sector tested in this paper is one component of a broader two-field EFT that may connect several open problems in cosmology. The same pNGB vacuum field χ that generates the conformal coupling also drives gauge-friction inflation (Section 4) and sources $\mathrm{U}(1)_{B\text{-}L}$ Chern-Simons baryogenesis (Section 6). The dark SU(2) gauge bosons that mediate the inflationary friction simultaneously produce, via adiabatic renormalization, a running vacuum coefficient $|\alpha_{\rm eff}| \sim 10^{-3}$ (Section 7).

These connections are *motivations*, not proven results. The baryogenesis chain yields η_B in the correct order of magnitude but carries $\mathcal{O}(1\text{–}10)$ uncertainty pending lattice validation. The running vacuum estimate relies on a Proca-reconstructed transverse coefficient whose independent derivation remains open. The inflationary sector is presented at the level of an effective action, without a detailed slow-roll analysis. Each of these components constitutes a separate research program; the present paper provides the common EFT architecture and tests the one sector—the Mass-Kick—that is most directly confrontable with current data.

12.4 Research Program

1. **Priority 1 (free f_φ + weak lensing):** Extend the MCMC to free f_φ and include DES Y3/KiDS-1000 weak lensing likelihoods, where the σ_8-reduction from the Mass-Kick provides the strongest discriminating power. This determines whether the Mass-Kick *improves* the fit relative to ΛCDM, not merely whether it is allowed.
2. **Priority 2 (Bayesian evidence):** Run nested sampling (`PolyChord`) for a definitive $\Delta \ln Z$ model comparison against ΛCDM and standard EDE.
3. **Priority 3 (Jordan-frame DM species):** Implement coupled DM as a separate CLASS species evolving on $\tilde{g}_{\mu\nu}$ with its own continuity and Euler equations. This may produce genuinely different k-dependence in the P(k) suppression and relax the S_8–Lyman-α trade-off.
4. **Priority 4 (baryogenesis transport):** Replace the compact efficiency formula with a full transport calculation including lattice validation.
5. **Priority 5 ($\alpha_{\rm eff}$):** Derive $c_T = -6$ independently via the projected Lichnerowicz heat kernel (a_2 coefficient for transverse spatial 1-forms), without using $c_{\rm Proca} = -13$ as input.

Formulate the dark SU(2) sector with explicit symmetry breaking and verify the mode counting in full curved-spacetime gauge-fixing.

13 Conclusions

We presented a two-field EFT framework whose Mass-Kick sector—a conformal coupling generating time-dependent dark matter mass—was implemented in `CLASS_EDE` and tested against cosmological data. The broader inflationary, baryogenesis, and running-vacuum connections of the framework remain programmatic (Section 12). The model reduces to GR + ΛCDM in the decoupling limit ($\Gamma \to 0$, $\lambda_g \to 0$, $\varphi \to 0$).

Two central results emerge:

(i) Boltzmann-code implementation. The first implementation of the conformal Mass-Kick coupling in `CLASS_EDE`, developed through three successive versions (friction-only → quasi-static screening → full Jordan-frame with CDM backreaction + fifth force), confirms that the mechanism suppresses σ_8 with correct sign and clean linear scaling $\Delta\sigma_8/\sigma_8 \approx -40\% \times \Gamma \times f_\varphi$.

(ii) MCMC constraint on Γ. A 10-parameter fit against Planck 2018+BAO+SH0ES (6 chains, $R-1 < 0.10$, $f_\varphi = 0.20$ fixed) yields $\Gamma = 0.062 \pm 0.043$ with $P(\Gamma > 0.01) = 90\%$ and a 95% upper limit $\Gamma < 0.14$. The posterior reveals a correlated EDE+Mass-Kick mode ($H_0 \approx 71$, $f_{\rm EDE} \approx 0.10$, $\Gamma \approx 0.03$) that is viable but subdominant to the ΛCDM-like solution. All six chains converge to $\Gamma \sim 0.06$, supporting the stability of the exploratory preference for $\Gamma > 0$.

Additionally: (i) a semi-analytic baryogenesis calculation yields η_B in the correct order of magnitude for a plausible parameter region, pending lattice validation; (ii) the operator structure, mode decomposition, and heat kernel a_1 coefficients for the dark SU(2) gauge sector are computed explicitly, yielding $|\alpha_{\rm eff}| \sim 3 \times 10^{-3}$ with transverse curvature enhancement ($a_{1,T}/a_{1,L} = 5/2$) compatible with the known Proca result $c_{\rm Proca} = -13$. The exact transverse coefficient $c_T = -6$ is reconstructed from the Proca constraint; its independent derivation via the projected Lichnerowicz heat kernel is deferred to future work.

The decisive next steps are: freeing f_φ in the MCMC, including weak lensing data (DES Y3, KiDS-1000) where the σ_8-reduction from the Mass-Kick provides the strongest discriminating power, and computing Bayesian evidence ($\Delta \ln Z$ vs. ΛCDM) via nested sampling.

Code availability: The modified `CLASS_EDE` source (`background.c` and `perturbations.c`, all three versions), Cobaya YAML configurations, and analysis scripts are available upon request.

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